



# ENGIN 604 INTRODUCTION TO PYTHON FOR FINANCE — FALL 2021

## HOMEWORK 1 - SOLUTION

**Due:** 11:59pm, Saturday, March 27

**Submit to:** [engin604assignments@gmail.com](mailto:engin604assignments@gmail.com)

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The file `returns_portfolios.pkl` contains a nested list that stores the monthly return of two portfolios ( $j = 1, 2$ ) from 2019-01-01 to 2020-12-01, plus an annualized risk-free rate in percentage terms with the same frequency. Using this file, answer the following:

1. Import the file `returns_portfolios.pkl` as shown below:

```
# imports the pandas library
import pandas as pd

# reads pickle
returns_portfolios = pd.read_pickle('returns_portfolios.pkl')
```

2. Compute the average return of each portfolio, defined as:

$$r_j = \frac{\sum_{i=1}^n x_i}{n}$$

```
total_ret_a = 0
total_ret_b = 0

for i, port in enumerate(returns_portfolios):
    for row in port:
        if i == 0:
            total_ret_a += row[1]
        else:
            total_ret_b += row[1]

mean_a = total_ret_a / len(returns_portfolios[0])
mean_b = total_ret_b / len(returns_portfolios[1])
```

3. Compute the standard deviation of each portfolio, defined as:

$$\sigma_j = \sqrt{\frac{\sum_{i=1}^n (x_i - r_j)^2}{n - 1}}$$

```
desv_squared_port_a = 0
desv_squared_port_b = 0
```

```

for j, port in enumerate(returns_portfolios):
    for row in port:
        if j == 0:
            desv_squared_port_a += (row[1] - mean_a) ** 2
        else:
            desv_squared_port_b += (row[1] - mean_b) ** 2

std_a = (desv_squared_port_a / (len(returns_portfolios[0]) - 1)) ** (1/2)
std_b = (desv_squared_port_b / (len(returns_portfolios[1]) - 1)) ** (1/2)

```

4. Compute the average of the risk-free rate. Since the risk-free rate is annualized, you must divide it by 12 to make it monthly and be consistent with the monthly return of the portfolios.

```

monthly_rfree = 0

for k, port in enumerate(returns_portfolios):
    for row in port:
        if k == 0:
            monthly_rfree += row[2] / 1200
        else:
            break

mean_rfree = monthly_rfree / len(returns_portfolios[0])

```

5. Compute the Sharpe ratio for each portfolio, defined as:

$$SR_j = \frac{r_j - r_f}{\sigma_j}$$

Where  $r_j$  is the average return of portfolio  $j$ ,  $r_f$  the average of the risk-free rate not in percentage terms and  $\sigma_j$  the standard deviation of portfolio  $j$ .

```

sr_a = (mean_a - mean_rfree) / std_a
sr_b = (mean_b - mean_rfree) / std_b

```

6. Check whether the following statements are true:

- a.  $SR_1$  is greater than  $SR_2$ .

```
sr_a > sr_b
```

```
## False
```

- b.  $SR_1$  is less than  $SR_2$ .

```
sr_a < sr_b
```

```
## True
```

- c. The difference between  $SR_2$  and  $SR_1$  is positive.

```
sr_b - sr_a > 0
```

```
## True
```

7. Print to the terminal the Sharpe ratio for each portfolio using the following format:

```
'The Sharpe ratio of portfolio j is {0:.4f}'
```

```

template_a = 'The Sharpe ratio of portfolio a is {0:.4f}'
template_a.format(sr_a)

```

```
## 'The Sharpe ratio of portfolio a is 0.4039'
```

```
template_b = 'The Sharpe ratio of portfolio b is {:.4f}'  
template_b.format(sr_b)
```

```
## 'The Sharpe ratio of portfolio b is 0.4660'
```